

work in many domains but the solar is still an infant and cannot carry them."

Pay package

Going by the data provided by *Times-Jobs.com*, package for fresher ranges from ₹ 120,000 to ₹ 180,000 per annum, and experienced professionals get around ₹ 300,000 to ₹ 420,000 per annum. The top roles for jobs are in the engineering and IT sectors, with Bengaluru being a hub of current job demand for junior and mid-level profiles, with starting salaries up to ₹ 500,000.

However, Advait Kumar informs us that this scale is not fixed everywhere. "It is important to understand that each engineering discipline is very different, and packages for freshers in engineering depend upon a candidate's area of concentration and level of education. I would say that for a fresher the package can range anywhere from ₹ 150,000 to ₹ 500,000 per annum."

Commenting on the pay package, Reddy says, "When it comes to me, I would prefer to take fresh diploma holders and fresh engineers with a pay pack of ₹ 10,000 and ₹ 15,000 per month as a start off in R&D. I believe in training them to the field requirements of designs right from the conceptualisation to the completely shaped product. This helps me in making my design engineers professionally ready to handle new requirements single handedly."

Skills expected by the industry

Basically, the engineers must be strong in power electronics. A fresh engineer should be well versed with electric circuits (network analysis) and solid-state electronics, and knowledge of advanced power semiconductor devices such as IGBTs, power MOSFETs and UJTs is most beneficial.

A fresher should have an exposure to basic designs of power supplies and power controllers. Understanding of DC-DC converters and recent trends in power conversion is a must. A regular update of the kind of PWM controllers' availability, and an understanding of

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new concepts in power switches could be of added advantage.

Similarly, knowledge in concepts of charging algorithms of different types of batteries will take him to a right place in R&D.

Reddy says, "I suggest experienced graduates to concentrate on power distribution, SCADA and electromechanical designs of the domain. As of today, most of the companies are importing the solar tracking systems. This is a simple application and can be indigenously developed."

He further adds, "Experienced engineers need to work on upgradation of imported ones with respect to basic functions, features, sustainability to climatic conditions and cost control. And, finally, they should be versatile for the adoption of technologies needed to cater to the product."

"Having the knowledge of electromagnetic compatibility and associated methods to combat it will help in staying compliant with industry standards, which becomes essential while dealing with high-frequency switching power circuits," informs Shankar.

Future

With ever-rising fuel costs, changing climatic conditions and an ever-growing demand for electricity, renewable energy resources like solar power are becoming an increasingly valuable part of the world's energy mix.

By utilising our expertise in power electronics, India needs to constantly strive and innovate to develop solutions such as solar inverters, low-voltage and grid-connection products

as well as PV power plants that can efficiently transform the sun's energy into reliable power.

Talking about power conditioning units (PCUs), Deyan says, "It holds great potential in the coming few decades for the Indian solar industry, and that is why we make so much investment and R&D into this. It is a combination of MPPT charge controller and pure sinewave inverter."

For controlling the cost of a solar products an engineer needs to put the SPV panel to full utility under all conditions in the design stage itself. Today, SPV modules are the backbone of the end product. But the scenario has to change to electronics playing a lead role.

In this regard, Reddy informs, "Updates till date that have come in include MPPT charger. MPPT is an algorithm catering to power for longer period, ultimately reducing the SPV module size and power. This reduces the cost of the SPV module. But, overall, product cost remains the same." So what is the advantage of MPPT charger with respect to the cost? "I suggest the fresh engineers to churn out new thoughts like boost regulating the power at each SPV module and then stretching its power to the junction box, which could be an answer to that question," he adds further. So here the skills of power electronics engineer take the lead role.

The current pay packages might not be very promising, but the solar industry has a big future. The window is open for the next 50 or more years. ●

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